

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

***CO1:** Apply the fundamental concepts of image formation, pinhole to perspective camera models, homogeneous coordinates, and multi-view geometry and interpret real-world visual scenes using OpenCV or MATLAB. (BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 20 Total Marks:40 Annexure A

SHORT ANSWER TEST

Code of Conduct:

I K.Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Marks
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding	

Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer	
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application	
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response	
Clarity & Presentation	10	Clear, concise, and well organized	Understandable with minor issues	Somewhat unclear or poorly	Difficult to understand	

		answer		structured		
--	--	--------	--	------------	--	--

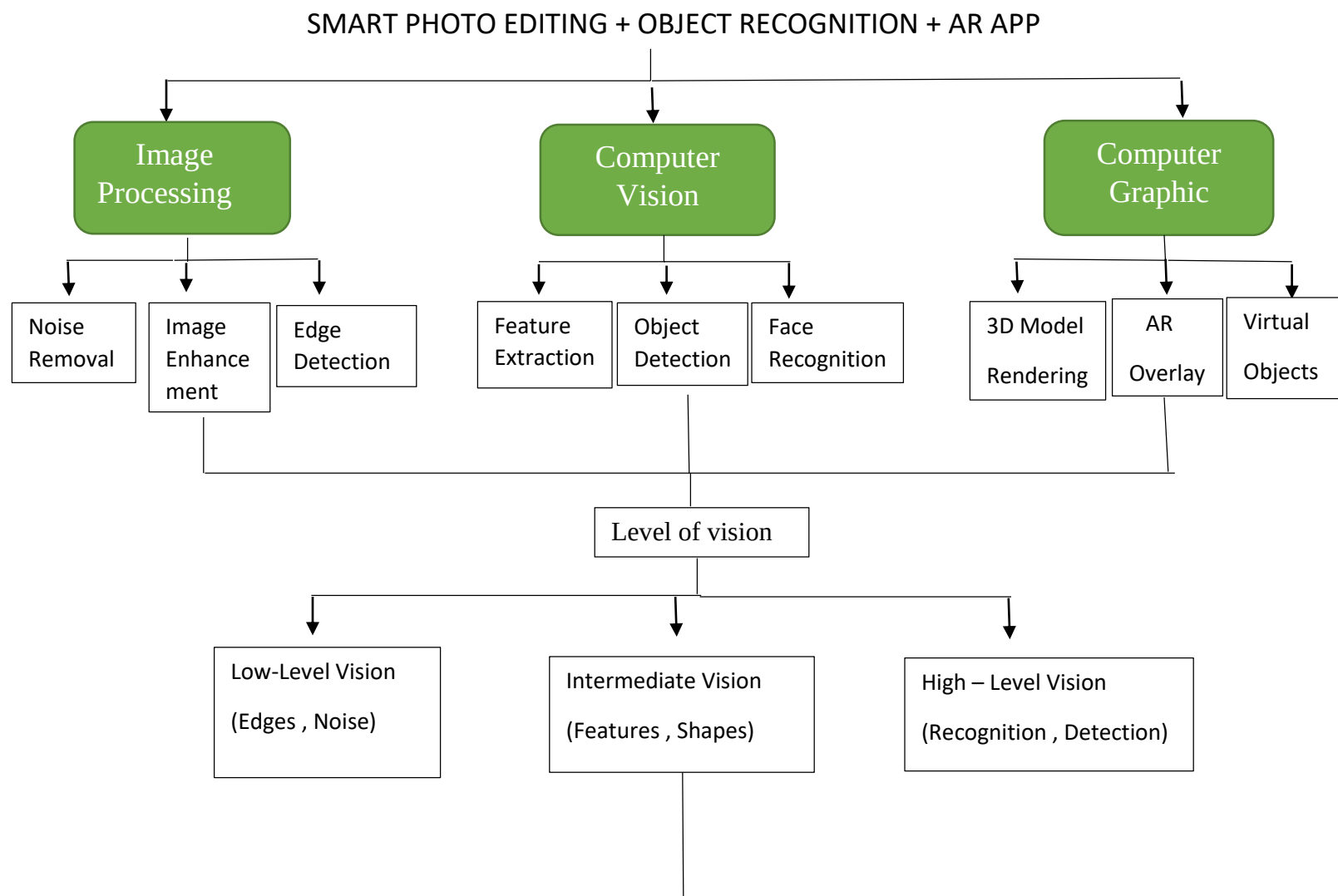
1. A company is developing a smart photo editing + object recognition + AR app.

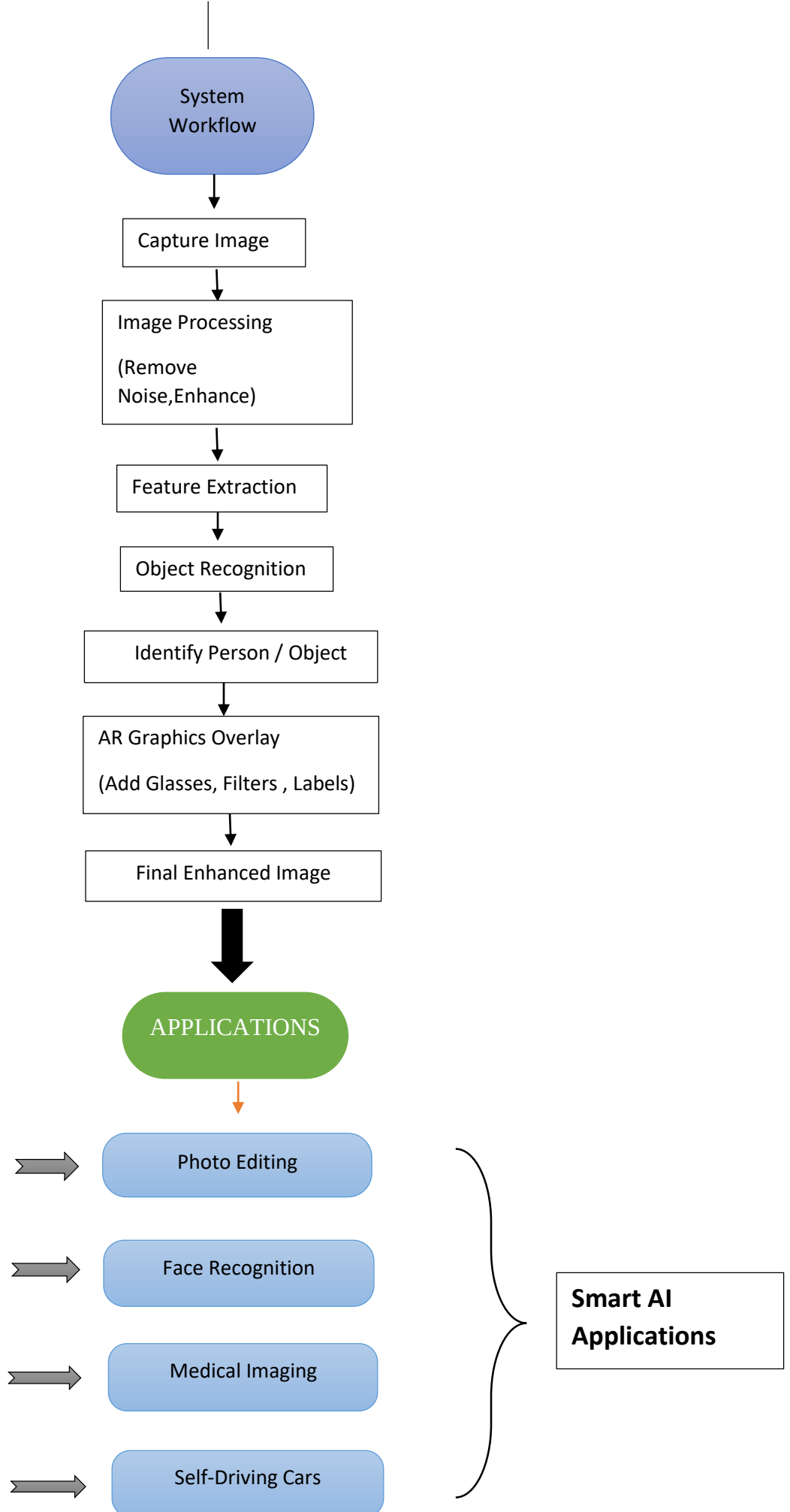
Create a concept map connecting:

- Image Processing
- Computer Vision
- Computer Graphics
- Levels of Vision
- Applications

Show:

- Workflow of the system
- Interconnections between components
- Real-world examples





Explanation

A Smart Photo Editing + Object Recognition + AR application combines Image Processing, Computer Vision, and Computer Graphics.

- **Image Processing** improves image quality by removing noise, enhancing brightness, and detecting edges.
- **Computer Vision** extracts features, recognizes objects, and identifies people using different **levels of vision**:
 - **Low-Level Vision**: Image enhancement and edge detection.
 - **Intermediate-Level Vision**: Feature extraction and object detection.
 - **High-Level Vision**: Object recognition and decision making.
- **Computer Graphics** generates virtual objects and overlays them on real-world images using **Augmented Reality (AR)**.

Workflow:

Capture Image → Image Processing → Feature Extraction → Object Recognition → AR Overlay → Final Enhanced Image

Real-world Examples:

- Snapchat and Instagram filters
- Google Lens
- Pokémon GO
- Medical image analysis
- Self-driving cars

